A 28nm Value-Wise Hybrid-Domain Compute-In-Memory Macro with Heterogeneous Memory Fabric and Asynchronous Sparsity Manager

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Edge AI devices need to be energy-efficient and compact while maintaining sufficient accuracy. Compute-in-memory (CIM) is a promising approach to achieve energy-efficient multiply-and-accumulate (MAC) operations. Analog CIMs (ACIMs) [1-5] generally offer higher energy efficiency but are sensitive to PVT variations, while digital CIMs (DCIMs) [6, 7] provide accurate computations but with mediocre energy efficiency. Combining analog and digital computing can help balance accuracy with power and area overheads. However, designing an energy-efficient hybrid CIM faces multiple challenges, as shown in Fig. 1. First, prior hybrid CIMs [8-10] still exhibit restricted efficiency due to their inherent bit-wise hybrid (BWH) structure, for example, accurate computation to the '0' value of most significant bits (MSBs) wastes energy. Second, DRAM-CIM [4], [5] typically demonstrates higher efficiency compared to SRAM-CIM [1-3], but it necessitates frequent off-macro refreshes, leading to additional system power and latency. Third, high bit-sparsity (>90%) in MSBs can significantly cause redundant MAC and quantization power.

This work addresses the above challenges using three key techniques: 1) a value-wise hybrid (VWH) structure that only accurately computes significant values, improving efficiency by ~2x while providing higher accuracy; 2) a heterogeneous memory (HM) fabric that combines the benefits of SRAM-CIM and DRAM-CIM; and 3) an asynchronous sparsity manager (ASM) that enables sparsity-adaptive throughput and power. Fabricated using a 28nm CMOS process, the proposed VWH-CIM macro achieves a competitive energy efficiency of 470.5 TOPS/W in 4b mode, 109.8 TOPS/W in 8b mode, and 48.5 TOPS/W in 12b mode.

Fig. 2 presents the overall architecture of the proposed VWH-CIM macro. It consists of 12× VWH cores, an ASM-embedded input driver, WL and weight drivers, a top controller, and a hierarchical results combiner (HRC). The VWH core includes an ACIM bank for the efficient computing of small values (inputs/weights), a DCIM bank for the precise computation of large weights, a digital MAC (DMAC) bank for the precise computation of large inputs, and a value dispatcher (VD) to allocate inputs to these three banks. The ACIM bank consists of a DRWL driver converting 2b-input into a pulse width per cycle, 30× HM units, and two ADCs. Each HM unit includes an 8×4b SRAM array and two 4b DRAM CIM cells, with the DRAM CIM cell split into 1b MSB and 3b LSBs to achieve adaptive signal margin. The weights of ACIM are converted into the current magnitudes of the DRAM CIM cells, acting on the outputs of the analog MAC, together with the pulse widths from the DRWL. The DCIM bank contains two units, each with an 8×4b SRAM array, a local 4b SRAM for stationary storage, and a 2b×4b digital process element (PE) to compute the large weights accurately. Similarly, two digital PEs in the DMAC are used to compute the large inputs accurately. The VWH-CIM macro can operate in three modes: 4-bit, 8-bit, and 12-bit, requiring 1, 2, and 3 VWH cores for weight storage, respectively. Compared to traditional DRAM-CIM mapping [4], [5], the HM-unit strategy prioritizes accuracy for the MSBs and efficiency for the rest, maximizing the balance between efficiency and accuracy within the ACIM. Thanks to data sparsity, a 5b-ADC and an 8b-ADC are sufficient to enable full-precision quantization for the MSB and LSB columns in the ACIM by allocating the LSB of the ADC equal to the LSB of the MAC, achieving an output ratio of 1, even using low-precision ADCs.

Fig. 3 (top) details the operations of the VWH scheme with enhanced computational efficiency and accuracy. Its computation is distributed to either the digital or analog domain, depending on the value of weight and input. During the preload phase, the initial 32 weights are analyzed to find the two most significant values, which are then loaded into the DCIM, with their indexes stored in the VD. The remaining 30 weights are loaded into the ACIM in their original order. During the MAC phase, two inputs are directly selected and sent to the DCIM according to the generated indexes of the two largest weights found earlier. Subsequently, the other two largest inputs in the remaining 30 are identified and sent to the DMAC. The indexes are used to retrieve the corresponding weights from the SRAM array in the ACIM and send them to the DMAC. The remaining 28 inputs are fed into the ACIM in their original sequence for highly efficient computation. The input dispatching via the VD is pipelined alongside the MAC operations, accounting for 12.7% of the area and 8.1% of the power consumption of the designed macro, without introducing additional latency. This VWH scheme focuses digital computation on the truly significant values that require precise calculations rather than the entire MSB, which allows for a higher proportion of analog processing, enhancing efficiency while maintaining compute accuracy.

Fig. 3 (bottom) depicts the HM fabric combining SRAM and DRAM, which reduces update and refresh latency as well as the power consumption of CIM. In contrast to traditional SRAM-CIM with an SRAM array and a local computing cell (LCC) [2, 3], the HM brings in two improvements. First, a DRAM unit replaces the LCC for stationary-weight storage and computation, saving energy for reading the SRAM array during each MAC operation. Second, a self-refresh unit (SRU) efficiently refreshes the DRAM, eliminating the need for off-macro data transfer in DRAM-CIM. The SRU includes a 1b-DAC and a 3b-DAC for programming the DRAM unit, while their area overhead is negligible as the SRUs on the same row share the same reference voltage. During the CIM update phase, the system prioritizes writing to the same row of the SRAM array, followed by a direct update to the DRAM unit. Afterward, the DRAM unit performs MAC operations in a weight-stationary data flow, while the SRAM array updates other rows. The proposed VWH scheme alleviates the leakage issue in DRAM cells. In this work, every 1000 computation cycles only require one refresh operation, resulting in negligible refresh power and accuracy loss. Moreover, this scheme allows for simultaneous MAC and update operations, reducing system latency. The 6T-SRAM utilizes a dual-port structure with two bit-lines for reading and writing, facilitating the parallel operations of reading the VWH for DMAC and updating the CIM. Evaluations across different models show that the HM with SRU achieves a system speedup of over 1.7×.

Fig. 4 introduces the ASM, which effectively utilizes sparsity to enhance the energy and latency of the MAC. The time-based ADC in the VWH-CIM cascades a voltage-to-time interface (V2T) and a time-to-digital converter (TDC), showing inherent scaling power to the data sparsity [11], which is hard to implement in voltage-based ADCs [1]. Moreover, the time differences between the output of V2T (T_0-T_{23}) and the rising edge of CLK are directly related to the MAC results. Therefore, the ASM can detect the maximum among T_0-T_{23}, enabling it to terminate the MAC cycle early and initiate the next one asynchronously. The time-based maximum detection is achieved using a simple array of AND gates, resulting in negligible area and power overhead for the ASM. Multi-bit data is processed in a standard bit-serial manner with 2 bits handled each cycle. Then, the MSB cycle saves more significant power and latency than the LSB one, as most data clusters around zero. Each MAC cycle is further divided into three sub-phases: 1) the reset phase pre-charges $DRBL_{M/L}$ of the DRAM and initializes the ADC; 2) the compute phase converts the 2b input into a pulse width using the DRWL driver to control the discharging of the DRAM; and 3) the quantization phase performs the ADC quantization. It is crucial to reduce the power and latency of the ACIM, as it dominates the VWH core compared to the DCIM and DMAC. Fortunately, the ASM alleviates this challenge by disabling redundant quantization and MAC in the ACIM. Evaluations of different models show that the ASM-assisted time-based ADC saves >47% power and reduces >72% latency.

Fig. 5 presents the evaluation and measurement results of the 28nm CMOS fabricated VWH-CIM. Compared to prior hybrid CIMs [8-10], the VWH achieves a sufficiently low MAC error with the highest hybrid efficiency (~90%), making it a promising hybrid structure. The VWH-CIM operates at voltages ranging from 0.6V to 1.0V, achieving the best energy efficiency at 0.6V and the best area efficiency at 1.0V. Benefiting from the ASM, the VWH-CIM demonstrates significant scaling capabilities in efficiency with sparsity. Under a condition of 90% input sparsity and 50% weight sparsity, a common scenario for unsigned inputs and binary complement weights, the energy efficiency scales from 470.5 to 87.1 TOPS/W in 4b mode, 109.8 to 25.1 TOPS/W in 8b mode, and 48.5 to 8.97 TOPS/W in 12b mode. Simultaneously, the area efficiency scales from 4.5 to 30.8 TOPS/mm^2 in 4b mode, 1.09 to 7.47 TOPS/mm^2 in 8b mode, and 0.35 to 2.43 TOPS/mm^2 in 12b mode.

Fig. 6 compares the VWH-CIM with state-of-the-art designs using full-analog CIMs, full-digital CIMs, and hybrid CIMs. This work advances beyond the efficiency bottlenecks of prior hybrid CIMs through a VWH structure. It combines the benefits of the highly efficient DRAM-CIMs and the refresh-free SRAM-CIMs using a heterogeneous memory fabric. A time-based ADC with the ASM reduces MAC power and latency by leveraging sparsity. With these techniques, the proposed VWH-CIM achieves a highly competitive overall FoM (energy efficiency × area efficiency). Moreover, it supports in-macro self-refresh and simultaneous MAC and update operations, offering significant advantages for system-level evaluations. The adaptive signal margin in the DRAM unit, in conjunction with the VWH, allows for more precise coordination of accuracy and efficiency. Fig. 7 presents the die photo, chip summary, and evaluations across different models. The VWH-CIM demonstrates highly competitive inference accuracy in classification, detection, and segmentation tasks, with all models exhibiting less than 1% accuracy loss.

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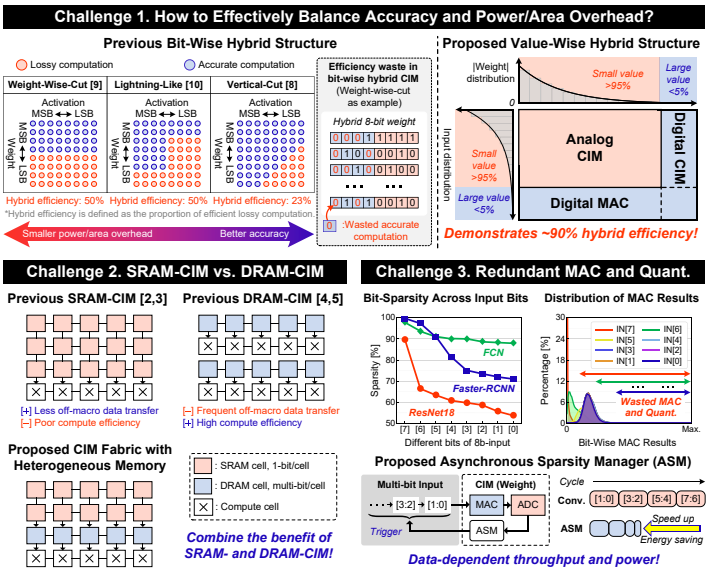


Fig. 1. Challenges of hybrid CIM macro and proposed technical features.

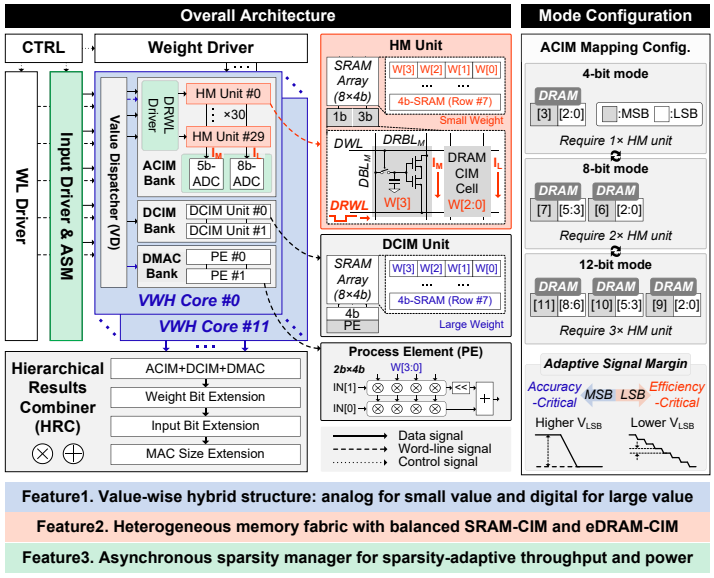


Fig. 2. The overall architecture of the proposed VWH-CIM macro.

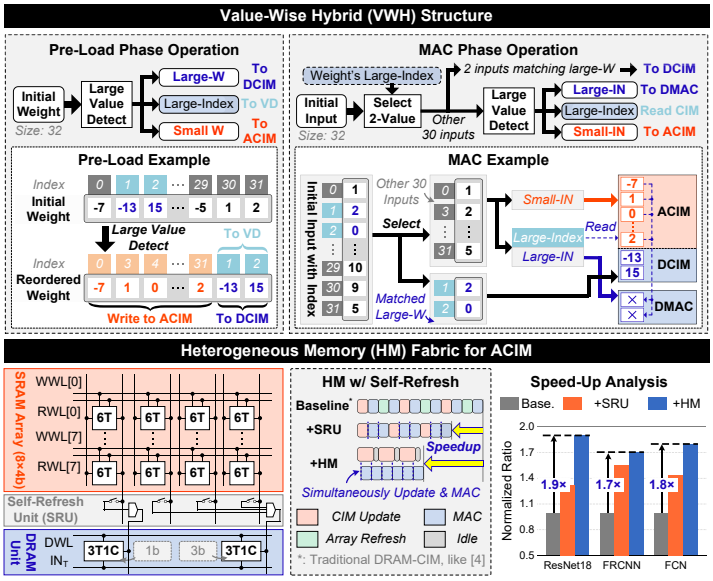


Fig. 3. Details of value-wise hybrid structure and heterogeneous memory fabric.

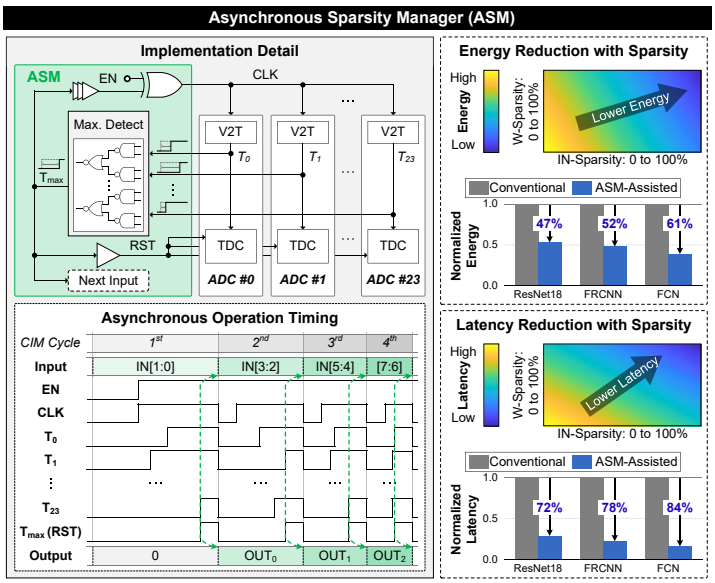


Fig. 4. Details of asynchronous sparsity manager.

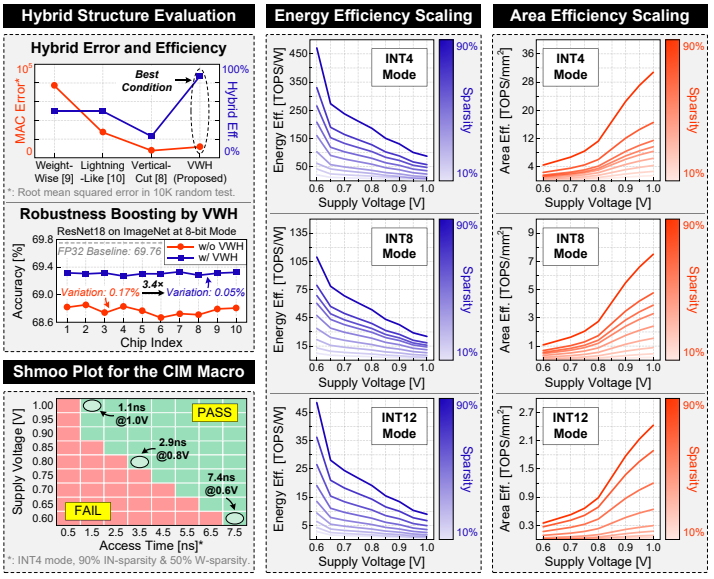


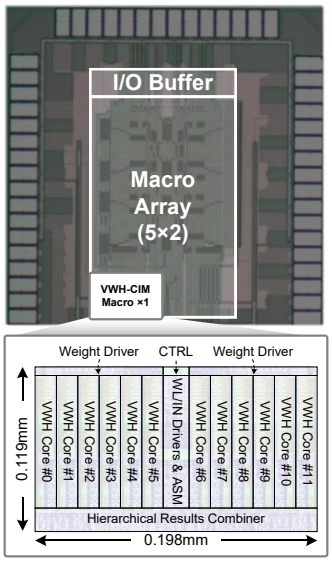
Fig. 5. Measurement results of the proposed VWH-CIM macro.

Comparison	Full-Analog CIM		Hybrid CIM			Full-Digital CIM	
Metrics	ISSCC'23 [1] S.-E. Hsieh	JSSC'24 [5] J. Song	VWH-CIM This Work	ISSCC'23 [8] P.-C. Wu	ISSCC'24 [10] A. Guo	ISSCC'21 [6] Y.-D. Chih	ISSCC'22 [7] D. Wang
Technology [nm]	12	65	28	22	22	22	28
Macro Area [mm²]	/	/	0.023	/	0.119	0.202	0.033
Supply Voltage [V]	0.5/0.85	/	0.6-1.0	0.6-0.8	0.6-0.9	0.52-0.92	0.5-1.1
Compute Method	Charge (Full-Analog)	Current (Full-Analog)	**Value-Wise Hybrid**	Vertical-Cut (Bit-Wise Hybrid)	Lightning-Like (Bit-Wise Hybrid)	Adder Tree (Full-Digital)	Appr. Adder (Full-Digital)
CIM Fabric	SRAM-CIM	DRAM-CIM	**Heterogeneous (SRAM + DRAM)**	SRAM-CIM	SRAM-CIM	SRAM-CIM	SRAM-CIM
Sparsity Technique	No	No	**Asynchronous Trigger**	Skipping	No	No	No
Precision Support	8b	4b	**4b/8b/12b**	BF16	8b	IN: 1-8b W: 4/8/12/16b	1b
Energy Efficiency [TOPS/W]	70.85-86.27 (8b)	233-304 (4b)	**470.5-87.1 (4b)[1] 109.8-25.1 (8b) 48.5-8.97 (12b)**	62.14-70.21 (BF16)	20.7-60.8 (8b)	11.7 (8b, 0.92V)	1108 (1b, 0.9V)
Area Efficiency [TOPS/mm²]	/	0.296 (4b)	**4.5-30.8 (4b)[1] 1.09-7.47 (8b) 0.35-2.43 (12b)**	/	0.524 (8b)	8.3 (8b, 0.92V)	278.5 (1b, 0.9V)
8b FoM[2]	/	4.31-5.62	**205.8 (0.9V)**	/	31.9	97.1 (0.92V)	75.5 (0.9V)
In-Macro Self-Refresh	/	No	**Yes**	/	/	/	/
Simultaneously MAC + Update	No	No	**Yes**	No	No	No	No
Hybrid Efficiency[3]	/	/	**87.50%**	23.40%	50%	/	/

1): Evaluated with 90% input sparsity and 50% weight sparsity. Efficiency scales with varied voltage.
2): FoM = energy efficiency × area efficiency. Normalized to 8b for fair comparison.
3): Hybrid efficiency is defined as the proportion of efficient lossy computation.

Fig. 6. Comparison table.

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Chip Summary			
Technology	28nm CMOS		
Macro Area	0.023mm^2		
Supply Voltage	0.6-1.0V		
# of Input Channels	32		
# of Output Channels	12		
Accumulation Size	32-2048		
$LSB_{Quantize}/LSB_{MAC}$ of ACIM	1.0		
Data Precision	4b	8b	12b
Output Precision	13-19b	21-27b	29-35b
Peak[1] Area Efficiency	30.8TOPS/mm^2 (4b)		
	7.47TOPS/mm^2 (8b)		
	2.43TOPS/mm^2 (12b)		
Peak[2] Energy Efficiency	470.5TOPS/W (4b)		
	109.8TOPS/W (8b)		
	48.5TOPS/W (12b)		

1): Evaluated at 1.0V, 90% input sparsity, 50% weight sparsity.
2): Evaluated at 0.6V, 90% input sparsity, 50% weight sparsity.

Evaluation on Different Models				
Model	VGG11	ResNet18	FRCNN	FCN
Task	Classification		Detection	Segment.
Dataset	Cifar10	ImageNet	COCO	VOC2012
Precision	4bit	8bit	12bit	12bit
Infer. Accuracy [%]	91.75 (Top1)	69.48 (Top1)	45.99 (mAP)	59.53 (MIoU)
Acc. Loss [%][3]	-0.15	-0.28	-0.73	-0.98

3): Compared with software baseline in FP32.

Fig. 7 Die photo, chip summary and evaluation on different models.

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